

Sai Udiga - CTF Platform Setup and Operations Guide

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System: ****Lloyds CTF / CTF-Main****

Repository: <https://github.com/saiudiga110/CTF-Main>

Local path: /Users/rislab/CTF-Main

1. Quick Answer

For normal event hosting, you should not need to configure the cluster manually again. From Mac mini 1, run:

```
cd /Users/rislab/CTF-Main
./scripts/ctfctl.sh start
```

Then open:

? Player site: <http://10.34.204.246>
? Admin: <http://10.34.204.246/admin>
? Architecture view: <http://10.34.204.246/plugins/ctfd-target/admin/architecture>
? Operations: <http://10.34.204.246/plugins/ctfd-target/admin/ops>

Use this to verify health:

```
./scripts/ctfctl.sh doctor
```

2. Current Machine Roles

Machine	Hostname	LAN IP	Role
Mac mini 1	rislab-mini1.local	10.34.204.246	Control plane + local worker
Mac mini 2	rislab-mini2.local	192.168.10.52	Remote Docker worker
Mac mini 3	rislab-mini3.local	192.168.10.53	Remote Docker worker

Mac mini 1 runs CTFd, MariaDB, Redis, proxy, Instance Manager, one local worker agent, and SSH tunnels to the remote worker agents. Mac mini 2 and Mac mini 3 run worker agents only. The tunnels are important on Docker Desktop because containers on Mac mini 1 may not route directly to the 192.168.10.x lab subnet.

3. Architecture

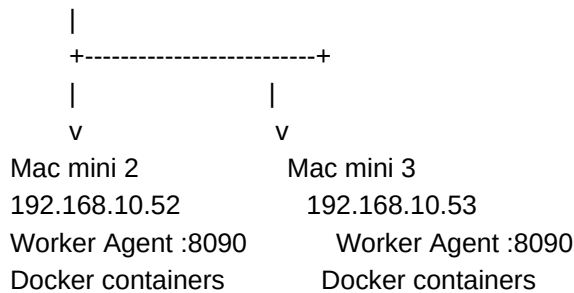
Players/Admin Browser

|
v

Mac mini 1 - 10.34.204.246

- Nginx proxy
- CTFd web application
- MariaDB
- Redis

- Instance Manager
 - Local Worker Agent
 - SSH tunnels:
- 127.0.0.1:18091 -> mac-mini-2:8090
 127.0.0.1:18092 -> mac-mini-3:8090



4. How Images Are Shared

Images are not pulled at player launch time. The startup command builds or verifies the required images on Mac mini 1 and then syncs current image IDs to Mac mini 2 and Mac mini 3.

Required workload images include:

- ? kali-ctf:latest
- ? vbank-ctf:latest
- ? vbank-auth:1
- ? vbank-idor:1
- ? vbank-logic:1
- ? vbank-xss:1
- ? vbank-staff-auth:1
- ? vbank-staff-deep:1
- ? vbank-analytics:latest

The startup script compares Docker image IDs. If a remote machine has an old or missing image, it sends a compressed Docker archive and loads it there. The same command also opens or reuses the SSH tunnels needed for the manager container to contact remote worker agents.

5. How Target Placement Works

When a player starts a target or Kali session:

1. The browser calls CTFd.
2. CTFd asks the Instance Manager on Mac mini 1.
3. The Instance Manager reads live worker heartbeats and capacity.
4. It chooses Mac mini 1, 2, or 3 based on available CPU/RAM/container/port capacity.
5. It tells that machine's worker agent to start the Docker container locally. For remote workers, the manager reaches the agent through the automatic SSH tunnel advertised as host.docker.internal:18091 or host.docker.internal:18092.
6. The selected worker returns a player URL like <http://192.168.10.52:20003/>.

If Mac mini 1 is full, new launches move to Mac mini 2 or Mac mini 3 automatically.

Target and Kali sidecars for the same player/challenge are colocated on the same worker so Kali can reach the target by vbank-app inside the shared Docker network.

6. Automation Commands

Run all commands from Mac mini 1:

```
cd /Users/rislab/CTF-Main
```

Start or resume everything

```
./scripts/ctfctl.sh start
```

Show status

```
./scripts/ctfctl.sh status
```

Show useful URLs

```
./scripts/ctfctl.sh urls
```

Run a full quick doctor check

```
./scripts/ctfctl.sh doctor
```

View local logs

```
./scripts/ctfctl.sh logs
```

Clean stale load-test containers and test records

```
./scripts/ctfctl.sh cleanup-tests
```

7. Resume or Start Next Time

Use this section when the Mac minis were restarted, Docker Desktop was quit, the lab was paused, or the LAN IP changed. You do not need to manually configure IP addresses, worker addresses, image sharing, or tunnels.

Run this only on Mac mini 1:

```
cd /Users/rislab/CTF-Main
```

```
./scripts/ctfctl.sh start
```

The resume command automatically performs these steps:

1. Detects the current Mac mini 1 LAN IP.
2. Updates .env with the detected HOST_IP.
3. Starts or resumes CTFd, proxy, database, cache, Instance Manager, and the local worker.
4. Connects to Mac mini 2 and Mac mini 3 over SSH.
5. Starts Docker Desktop on the remote machines if needed.
6. Syncs the latest repo code and workload images to the remote machines.
7. Creates or reuses SSH tunnels for remote worker control traffic.
8. Starts or restarts the remote worker agents.
9. Waits for worker heartbeats so the architecture page can show all machines.

After the command finishes, print the current URLs:

```
./scripts/ctfctl.sh urls
```

Then verify all three Mac minis are visible:

```
./scripts/ctfctl.sh status
```

Expected healthy output includes:

Cluster: OK

? mac-mini-1: fresh=True inventory=True

? mac-mini-2: fresh=True inventory=True

? mac-mini-3: fresh=True inventory=True

If Mac mini 1 receives a different LAN IP after reboot, use the new URL printed by `./scripts/ctfctl.sh urls`. The automation updates the application, but players still need the current address unless the router reserves a fixed DHCP address for Mac mini 1.

Recommended production setup:

- ? Reserve Mac mini 1, Mac mini 2, and Mac mini 3 IP addresses in the router DHCP settings.
- ? Keep Docker Desktop set to start automatically on all three machines.
- ? Run `./scripts/ctfctl.sh start` before every event or lab session.
- ? Keep the Architecture page open during the event.

Resume troubleshooting:

Symptom Fix

URL changed after reboot Run `./scripts/ctfctl.sh urls` and use the printed IP URL

Architecture page misses Mac mini 2 or 3 Run `./scripts/ctfctl.sh start` again to refresh SSH tunnels and worker heartbeats

Remote worker inventory is false Check SSH with `ssh rislab@rislab-mini2.local` hostname and `ssh rislab@rislab-mini3.local` hostname

Docker is not ready on a remote Mac Open Docker Desktop on that Mac, then rerun `./scripts/ctfctl.sh start`

Old crash-test containers remain Run `./scripts/ctfctl.sh cleanup-tests`, then

```
./scripts/ctfctl.sh status
```

8. Installation From GitHub on a Fresh Mac mini 1

```
cd /Users/rislab
git clone https://github.com/saiudiga110/CTF-Main.git
cd CTF-Main
chmod +x start.sh scripts/start-macmini-cluster.sh scripts/ctfctl.sh
./scripts/ctfctl.sh start
```

Prerequisites:

- ? Docker Desktop installed and running on all three Mac minis.
- ? Passwordless SSH from Mac mini 1 to Mac mini 2 and Mac mini 3.
- ? Same LAN connectivity between the three machines.
- ? SSH local forwarding allowed from Mac mini 1 to each remote Mac mini; ctfctl start creates the required forwards automatically.
- ? Mac mini 2 hostname: rislab-mini2.local.
- ? Mac mini 3 hostname: rislab-mini3.local.

9. Available Endpoints

User and player endpoints

Endpoint	Purpose
/	CTFd player site
/challenges	Challenge list
/plugins/ctfd-target/target/start	Start player target
/plugins/ctfd-target/target/status	Target status
/plugins/ctfd-target/target/stop	Stop target
/plugins/ctfd-target/kali/start	Start Kali/noVNC sidecar
/plugins/ctfd-target/kali/status	Kali status
/plugins/ctfd-target/kali/sso-url	Kali noVNC access URL
/plugins/ctfd-target/kali/stop	Stop Kali sidecar
/plugins/ctfd-target/health	Public plugin health

Admin endpoints

Endpoint	Purpose
/admin	CTFd admin panel
/plugins/ctfd-target/admin/architecture	Live architecture view
/plugins/ctfd-target/admin/architecture.json	Architecture JSON for UI refresh
/plugins/ctfd-target/admin/ops	Event operations dashboard
/plugins/ctfd-target/admin/network	Network and DNS settings
/plugins/ctfd-target/admin/settings	Target/Kali settings
/plugins/ctfd-target/admin/dashboard	Running instance dashboard

Instance Manager endpoints

Endpoint Purpose

http://10.34.204.246:8088/health Manager health
/workers/heartbeat Worker registration heartbeat
/nodes Current fresh workers
/architecture Full architecture and inventory data
/instances Instance allocation/listing
/instances/<id>/status Instance status
/instances/<id> DELETE Stop instance

10. Production Notes

? The cluster is currently clean and healthy with 3 fresh nodes and zero test containers.
? Current image sizes are reduced: vBank about 170 MB and Kali about 2.85 GB.
? Architecture page auto-refreshes every 10 seconds.
? Worker heartbeats now retry after transient manager timeouts instead of exiting.
? Instance allocation is locked in the manager to avoid duplicate port assignment under concurrent launch bursts.
? DNS port 53 on Mac mini 1 is currently occupied by another local service. The app still works through direct IP: http://10.34.204.246.

10. Capacity Test Result

The current cluster reliably reached 21 Kali + 21 vBank target pairs in sequential testing before Docker create calls became too slow and timed out. The requested 30+ Kali sessions are close, but not yet reliable with on-demand Kali creation.

To make 30+ Kali sessions production reliable, use one or both of these improvements:

1. Pre-warm a pool of Kali containers before players arrive.
2. Reduce Kali startup cost further or split Kali sessions across more worker machines.

The automation is ready, but the platform should avoid sudden burst creation of 30+ Kali desktops at once unless a warm pool is implemented.

11. Capacity and Crash Test Result

The Kali image was reduced to a focused event profile while keeping the tools requested for the lab:

? Firefox browser
? XFCE terminal
? Burp Suite
? nmap
? Hydra and John password tools
? apt package installation support
? noVNC desktop access

Removed from the default image to save space:

- ? Kali themes and wallpapers
- ? hashcat by default
- ? full rockyou wordlist by default
- ? extra fuzzers/scanners such as sqlmap, ffuf, gobuster, wfuzz, nikto, dirb, tcpdump, proxychains, and mitmproxy by default

Final verified image size on Mac mini 1: about 1.14 GB for kali-ctf:latest.

Runtime defaults were also reduced so more sessions can fit on the three Mac minis:

Setting New default Purpose

KALI_MEM_LIMIT 1280m Enough for Firefox + Burp for lab use

KALI_CPU_QUOTA 50000 Half CPU request per Kali session

KALI_DISK_MB 1536 Lower per-session disk reservation

KALI_SHM_SIZE 256m Lower shared memory reservation

Optional build flags can add tools back when needed:

KALI_INCLUDE_EXTRA_TOOLS=1 docker compose build kali

KALI_INCLUDE_HASHCAT=1 docker compose build kali

KALI_INCLUDE_WORDLISTS=1 docker compose build kali

Earlier crash testing reached 21 Kali + 21 vBank target pairs with the older, heavier profile. The new profile should allow more sessions, but a fresh 30+ concurrent crash test should be run before a large event because Burp and Firefox still consume real RAM when actively used.

12. Troubleshooting

Check everything

```
./scripts/ctfctl.sh doctor
```

Architecture page does not show all 3 nodes

Run the startup command again. It refreshes worker containers, heartbeats, image sync, and SSH tunnels.

```
./scripts/ctfctl.sh start
```

Then wait 30 seconds and run:

```
./scripts/ctfctl.sh status
```

Remote worker is stale

Check SSH:

```
ssh rislab@rislab-mini2.local hostname  
ssh rislab@rislab-mini3.local hostname
```

Check tunnel listeners on Mac mini 1:

```
ls -l /dev/tcp/10.34.204.246/18091  
ls -l /dev/tcp/10.34.204.246/18092
```

Check Docker on remote:

```
ssh rislab@rislab-mini2.local 'docker ps'  
ssh rislab@rislab-mini3.local 'docker ps'
```

Then restart the whole cluster:

```
./scripts/ctfctl.sh start
```

Stale test containers remain

```
./scripts/ctfctl.sh cleanup-tests  
./scripts/ctfctl.sh status
```

DNS does not work

Use direct IP:

```
http://10.34.204.246
```

DNS port 53 is occupied on Mac mini 1. This does not block direct-IP operation.

Images are old on remote machines

Run:

```
./scripts/ctfctl.sh start
```

The startup script compares Docker image IDs and syncs changed images automatically.

13. Files Added or Changed for Operations

File Purpose

scripts/ctfctl.sh Operator command wrapper

scripts/start-macmini-cluster.sh Full cluster start/sync command

target-plugin/templates/architecture.html Architecture page UI

infra/instance_manager/app.py Manager scheduling, architecture API, allocation lock

infra/instance_manager/worker_agent.py Worker inventory and resilient heartbeat

infra/instance_manager/docker_worker.py Serialized Docker create operation
challenges/parrot/Dockerfile Reduced Kali image size
challenges/vbank-ctf/Dockerfile Reduced vBank image size

14. Daily Event Checklist

Before event:

```
cd /Users/rislab/CTF-Main  
./scripts/ctfctl.sh start  
./scripts/ctfctl.sh doctor
```

During event:

? Keep Architecture open: <http://10.34.204.246/plugins/ctfd-target/admin/architecture>
? Keep Operations open: <http://10.34.204.246/plugins/ctfd-target/admin/ops>

After event:

```
./scripts/ctfctl.sh cleanup-tests
```

Stop player instances from the Operations page if needed.